Employee Attendance & Analytics API

HROne Software Engineer Trainee — Project Summary

Project Overview

This project implements an employee attendance and analytics API using Python, FastAPI, and MongoDB. It covers employee punch-in/punch-out workflows, attendance corrections with an audit history, and analytics endpoints. The repository is intended to be submitted as a public GitHub repository, as specified in the HROne assignment email.

Technology Stack

Component	Technology / role
API framework	FastAPI (Python)
Database	MongoDB
API testing	Swagger UI / OpenAPI documentation
Version control	Git and GitHub

Implemented and Tested Behaviors

- Punch-out validation: an already-closed attendance record was tested and returned HTTP 409 Conflict after the duplicate punch-out handling change.
- Punch duration validation: a punch-out more than 24 hours after punch-in was tested and returned HTTP 422 Unprocessable Entity.
- Attendance pagination: the user reported that pagination and related endpoints were working in Swagger testing.
- Employee monthly analytics and department analytics: the user reported that these endpoints were checked successfully; detailed independent verification of every calculated value is not documented here.
- Manual attendance correction: a PATCH request returned HTTP 200 and produced an audit-history entry containing the editor, reason, timestamp, and changed field with from/to values.

Known Limitation / Follow-up

During testing, a test attendance record for EMPTEST01 had date 2026-10-12 while its punch-in and punch-out timestamps were on 2026-10-14. This mismatch produced an overtime value of 2730 minutes. The shift-end helper appeared reasonable on inspection, but the inconsistent record means the overtime result should not be treated as validated. The record and calculation should be checked using consistent attendance dates and timestamps. This summary does not claim that this issue has been fixed.

Older test-history entries also used a different format from newer entries. The latest correction entry used a from/to mapping, but historical entries were not normalized in the observed response.

Repository and Submission Notes

- Repository URL: https://github.com/ganreddy77/hrone-attendance-api
- The latest Git commit reported during the session was 93f6297 (Fix duplicate punch-out handling), and the working tree was clean and synchronized with origin/main at that point.
- HROne's email requires app/main.py, requirements.txt, REVIEW.md, DECISIONS.md, and README.md in a public repository.
- Do not include .env files, credentials, secrets, or a Dockerfile. Keep the repository public until 31 October 2026, as requested in the assignment email.

Note: This is a supplementary project summary. It does not replace the required REVIEW.md and DECISIONS.md files or the official HROne submission form.